TMLE Under the Hood

Technical supplement for Segment G

Navigator → nuisance fits → targeting → inference

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This Supplement Opens the Steps Hidden in the Live Demo

Live version

- One continuous Navigator-to-estimate story
- Compact `tmle()` call
- Core diagnostics and interpretation
- Script: `pneumonia_tmle_example.R`

Technical version

- Data generator and cohort verification
- Super Learner fits for Q and g
- Clever covariate and targeting regression
- Efficient influence function and interval
- Script: `pneumonia_tmle_technical.R`

Both versions use the same frozen cohort, treatment, outcome, adjustment set, estimand, and workshop runtime settings.

The Navigator Defines the Statistical Problem

Observed unit

$$O = (W, A, Y)$$

with $W = \{\text{age, prior pneumonia, prior vaccine}\}$.

Causal target

$$\Psi(P_0) = E_0\left[Y^{(1)} - Y^{(0)}\right].$$

Time order

1. Baseline history W
2. Baseline vaccination A
3. Pneumonia hospitalization Y by 12 months

The estimator is selected after this target and time order are fixed.

Identification Connects the Causal Target to Observed Data

Under consistency, conditional exchangeability, and positivity,

$$E\left[Y^{(a)}\right] = E_W\{E(Y \mid A = a, W)\}.$$

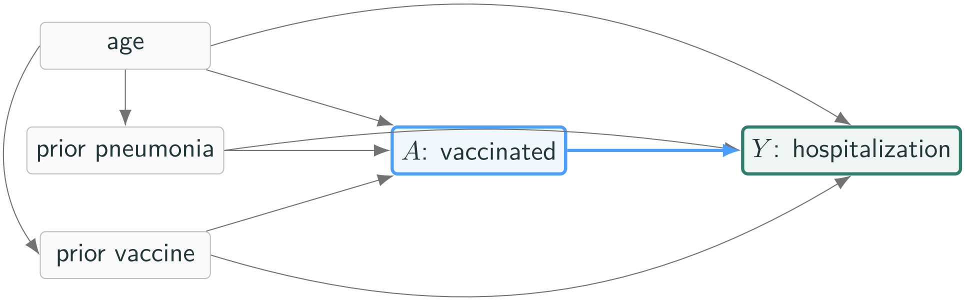
Therefore the identified risk-difference parameter is

$$\Psi(P_0) = E_W\{Q_0(1, W) - Q_0(0, W)\}, \qquad Q_0(a, w) = E(Y \mid A = a, W = w).$$

Division of labor

The causal assumptions identify a statistical parameter. TMLE estimates that parameter from the observed data distribution.

The simcausal DAG Includes Baseline History and Treatment Selection



Age affects both prior-history variables; all three baseline variables affect vaccination and outcome. Vaccination has a protective structural effect.

The Generator Creates Confounding by Indication

Vaccination mechanism

$$\begin{aligned}\operatorname{logit} P(A=1 \mid W) = &-1.40 + 0.25\,\frac{\text{age}-65}{10} \\ &+ 1.40\,\text{priorPneumonia} \\ &+ 1.20\,\text{priorVaccine}.\end{aligned}$$

Higher-risk histories increase vaccination probability.

Outcome mechanism

$$\begin{aligned}\operatorname{logit} P(Y=1 \mid A, W) = &-3.00 - 0.80A \\ &+ 0.25\,\frac{\text{age}-65}{10} \\ &+ 1.60\,\text{priorPneumonia} \\ &- 0.15\,\text{priorVaccine}.\end{aligned}$$

The treatment coefficient is protective on the log-odds scale.

The observational treatment groups therefore differ before vaccination.

The Teaching Cohort Is Fixed and Verifiable

One frozen synthetic cohort

- 5,000 adults
- 1,627 vaccinated (32.5%)
- 334 hospitalizations (6.7%)
- Shared unchanged by Modules F and G

Why freeze the data?

- Everyone analyzes the same rows.
- Results on these slides match what you will reproduce.
- The archived generator script rebuilds the cohort and verifies it row by row.

Integrity details — the file checksum and the generator's verification steps — are documented with the repository materials.

TMLE Uses Two Nuisance Functions and One Targeting Step

Q estimates conditional outcomes; g describes treatment assignment; the update solves the estimating equation associated with the selected risk-difference target.

The Initial Outcome Regression Learns $Q(A, W)$

$$Q_0(a, w) = E(Y \mid A = a, W = w).$$

Fit the outcome model using observed A, W, and Y:

$$\widehat{Q}(A_i, W_i).$$

These observed-treatment predictions are used in the targeting regression.

Predict each participant under both treatment strategies:

$$\widehat{Q}(1, W_i), \qquad \widehat{Q}(0, W_i).$$

These counterfactual predictions are updated and averaged.

Predictions are bounded away from 0 and 1 before applying the logit update.

Super Learner Builds the Initial Nuisance Fits

Workshop library

- `SL.glm`: main-effects logistic regression
- `SL.glmnet`: penalized logistic regression
- `SL.ranger`: random forest

Ensembling

- Each candidate generates cross-validated predictions.
- The metalearner combines them to minimize prediction risk.
- The workshop uses three folds to keep the live run practical.

Flexible prediction can reduce nuisance-model misspecification, while the causal target and identification assumptions remain fixed upstream.

G-Computation Is the Initial Plug-In Contrast

Before targeting,

$$\widehat{\Psi}_{\text{gcomp}} = \frac{1}{n} \sum_{i=1}^{n} \left\{ \widehat{Q}(1, W_i) - \widehat{Q}(0, W_i) \right\}.$$

In the transparent parametric comparator:

$$\widehat{\Psi}_{\text{gcomp}} = -3.28 \text{ percentage points.}$$

This estimate relies directly on the outcome model. TMLE retains the standardization step but adds information from the treatment mechanism.

The Treatment Mechanism Learns $g(W)$

$$g_0(w) = P(A = 1 \mid W = w).$$

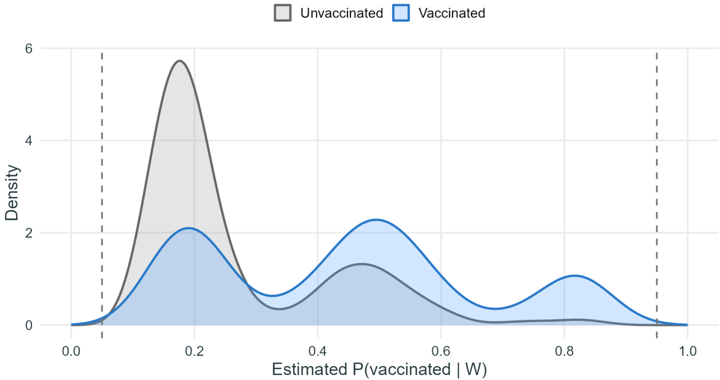
Role in diagnostics

- Describes treatment selection given measured history
- Reveals limited support and near-deterministic treatment
- Motivates redesign, restriction, or a revised target when needed

Role in estimation

- Defines inverse-probability weights
- Enters TMLE through the clever covariate
- Determines where outcome residuals receive greater leverage

The Observed Data Show Shared Treatment Support



Parametric diagnostic

Range:

$$0.077 \text{ to } 0.909$$

Share outside $[0.05, 0.95]$:

$$0\%.$$

The plot assesses empirical support across measured W; it does not test exchangeability.

The Clever Covariate Focuses the Update

For a binary treatment and the average risk difference,

$$H(A, W) = \frac{A}{\widehat{g}(W)} - \frac{1 - A}{1 - \widehat{g}(W)}.$$

Under the two intervention values:

$$H(1, W) = \frac{1}{\widehat{g}(W)}, \qquad H(0, W) = -\frac{1}{1 - \widehat{g}(W)}.$$

Observations receiving a treatment that was less probable given W contribute more strongly to the targeted residual correction.

A Logistic Fluctuation Estimates One Targeting Coefficient

Fit the intercept-free fluctuation model

$$\operatorname{logit} E(Y \mid A, W) = \operatorname{logit} \widehat{Q}(A, W) + \epsilon H(A, W),$$

where $\operatorname{logit} \widehat{Q}(A, W)$ is an offset.

In the manual decomposition:

$$\widehat{\epsilon} = 0.008503.$$

The update is low-dimensional and parameter-focused: the complex learning happened in the initial nuisance fits.

The Targeting Step Updates Both Intervention Predictions

$$\widehat{Q}^*(1, W) = \operatorname{expit}\left[\operatorname{logit}\widehat{Q}(1, W) + \widehat{\epsilon}\frac{1}{\widehat{g}(W)}\right],$$
$$\widehat{Q}^*(0, W) = \operatorname{expit}\left[\operatorname{logit}\widehat{Q}(0, W) - \widehat{\epsilon}\frac{1}{1 - \widehat{g}(W)}\right].$$

Because the update is on the logit scale, all targeted predictions remain in the valid probability range.

The TMLE Is a Plug-In Contrast of Updated Risks

$$\widehat{\Psi}_{\mathrm{TMLE}} = \frac{1}{n}\sum_{i=1}^{n}\left\{\widehat{Q}^*(1, W_i) - \widehat{Q}^*(0, W_i)\right\}.$$

Manual decomposition	**`tmle()` implementation**
-3.00 percentage points	-3.05 percentage points

Small differences reflect implementation details in nuisance fitting and targeting; both use the same data, adjustment set, and estimand.

The Efficient Influence Function Supplies Uncertainty

For the average risk difference,

$$D^*(O) = H(A,W)\{Y - Q^*(A,W)\} \\ + Q^*(1,W) - Q^*(0,W) - \Psi.$$

Estimate

$$\widehat{\mathrm{SE}}(\widehat{\Psi}) = \frac{\mathrm{sd}\{\widehat{D}^*(O_i)\}}{\sqrt{n}}, \qquad \widehat{\Psi} \pm 1.96\,\widehat{\mathrm{SE}}.$$

The manual implementation has $\overline{\widehat{D}^*} = -8.8 \times 10^{-13}$, a numerical check that the targeted estimating equation is solved. Its SE is 0.802 percentage points.

Manual Step 1: Fit Q and g

```r
learners <- c("SL.glm", "SL.glmnet", "SL.ranger")

Q_fit <- SuperLearner(
  Y = Y, X = data.frame(A = A, W),
  family = binomial(), SL.library = learners,
  cvControl = list(V = 3L)
)
QAW <- Q_fit$SL.predict
Q1  <- predict(Q_fit, data.frame(A = 1, W))$pred
Q0  <- predict(Q_fit, data.frame(A = 0, W))$pred

g_fit <- SuperLearner(
  Y = A, X = W, family = binomial(),
  SL.library = learners, cvControl = list(V = 3L)
)
g <- pmin(pmax(g_fit$SL.predict, 0.025), 0.975)
H <- A / g - (1 - A) / (1 - g)
```

Manual Step 2: Target, Average, and Form the Interval

```r
fluctuation <- glm(
  Y ~ -1 + H + offset(qlogis(QAW)),
  family = binomial()
)
epsilon <- coef(fluctuation)[["H"]]

Q1_star <- plogis(qlogis(Q1) + epsilon / g)
Q0_star <- plogis(qlogis(Q0) - epsilon / (1 - g))
QAW_star <- plogis(qlogis(QAW) + epsilon * H)

psi <- mean(Q1_star - Q0_star)
eif <- H * (Y - QAW_star) +
  Q1_star - Q0_star - psi
se <- sd(eif) / sqrt(length(Y))
ci <- psi + qnorm(c(0.025, 0.975)) * se
```

The Package Implementation Encapsulates the Same Sequence

```
fit <- tmle(
  Y = Y, A = A, W = W,
  Q.SL.library = learners,
  g.SL.library = learners,
  V.Q = 3, V.g = 3
)

psi <- fit$estimates$ATE$psi
ci  <- fit$estimates$ATE$CI

g1W <- fit$g$g1W
Q0_initial <- fit$Qinit$Q[, "Q0W"]
Q1_initial <- fit$Qinit$Q[, "Q1W"]
Q0_updated <- fit$Qstar[, "Q0W"]
Q1_updated <- fit$Qstar[, "Q1W"]
```

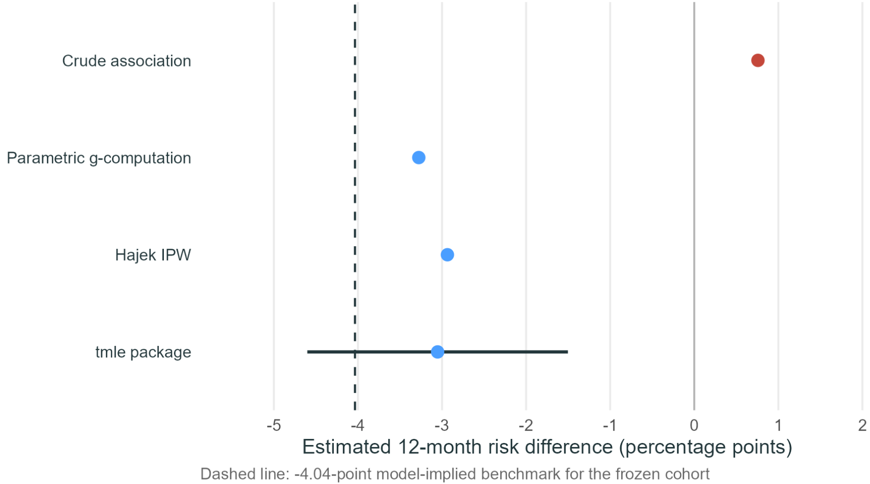
The manual version exposes the mechanics; the package version is the practical analysis route.

The Technical Results Keep Every Estimator on One Scale

Estimator	RD (pp)	95% lower	95% upper
Crude association	$+0.76$	–	–
Parametric g-computation	-3.28	–	–
Hajek IPW	-2.94	–	–
Manual TMLE decomposition	-3.00	-4.57	-1.43
`tmle()` package	-3.05	-4.60	-1.50
DGP benchmark (frozen W)	-4.04	–	–

The benchmark averages model-implied counterfactual risks over the frozen covariate distribution; it would not be observed in an applied analysis.

The Estimator Comparison Separates Association from Adjustment



The adjusted estimators recover a protective direction in this sample. Their similarity is useful for implementation review, while causal validity still depends on the shared study assumptions.

Double Robustness Uses Information from Both Nuisance Models

For a point-treatment average effect, TMLE is consistent under appropriate regularity and positivity conditions if either

$$\widehat{Q} \longrightarrow Q_0 \qquad \text{or} \qquad \widehat{g} \longrightarrow g_0.$$

Practical implication

- Fit both nuisance functions carefully.
- Inspect support and prediction behavior.
- Use flexible learners when justified.

Scope

- This is an estimator property.
- It does not identify unmeasured confounding.
- Finite-sample performance can still be poor.

Positivity Diagnostics Guide the Estimand and Analysis Population

When $\widehat{g}(W)$ approaches 0 or 1,

$$|H(A, W)|$$

can become large, increasing variance and sensitivity to nuisance-model error.

Analysis options

- Improve design or data collection
- Restrict to a population with shared support
- Use a treatment rule or overlap-weighted target

Truncation

- Can stabilize an estimator
- Introduces a bias-variance tradeoff
- Should be prespecified and sensitivity-analyzed

Influence-Curve Inference Requires Rate and Regularity Conditions

The usual Wald interval relies on an asymptotic linear expansion:

$$\widehat{\Psi} - \Psi_0 = \frac{1}{n}\sum_{i=1}^{n} D^*(O_i) + o_p(n^{-1/2}).$$

A key remainder is controlled when the product of nuisance errors is small:

$$\|\widehat{Q} - Q_0\| \, \|\widehat{g} - g_0\| = o_p(n^{-1/2}).$$

Cross-fitting can reduce empirical-process restrictions for highly adaptive learners. The workshop script uses the standard single-fit `tmle` implementation with cross-validated Super Learner nuisance fits.

The Causal Interpretation Retains the Navigator Assumptions

Identification

- Exchangeability given measured W
- Positivity in the target population
- Consistency of treatment versions

Measurement

- Correct vaccination classification
- Correct hospitalization ascertainment

Time and selection

- Common eligibility and treatment time zero
- Complete or appropriately modeled follow-up
- Appropriate handling of missingness and selection

Transport

- The target population is the synthetic cohort
- No clinical generalization is intended

The Audit Trail Connects Data, Code, Output, and Session State

Artifact	Role
`data/pneumonia_data.csv`	Frozen cohort used throughout the Navigator and Segment G
`simulate_pneumonia_data.R`	Structural equations, cohort regeneration, and row-by-row verification
`pneumonia_tmle_example.R`	Short live analysis
`pneumonia_tmle_technical.R`	Manual decomposition, package fit, diagnostics, plots, and outputs
`outputs/*.csv`	Machine-readable estimates and diagnostics
`outputs/session_info.txt`	R and package environment

Rebuilding begins with the technical R script, then compiles each Beamer source twice.

TMLE, AIPW, and DML Share an Orthogonal-Score Perspective

	TMLE	AIPW / one-step	DML
Core idea	Target an initial plug-in fit	Add an influence-function correction	Estimate an orthogonal score with sample splitting
Nuisance functions	Q and g	Q and g	Problem-specific nuisance functions
Parameter space	Plug-in estimate respects outcome bounds	One-step estimate need not be substitution-based	Depends on score and target
Inference	EIF-based under regularity/rate conditions	EIF-based under regularity/rate conditions	Orthogonal-score inference, typically cross-fitted

These are related frameworks, not interchangeable labels; implementation details and target parameters determine the exact guarantees.

Technical Takeaways

1. The Navigator fixes the population, time order, adjustment set, and risk-difference target before estimation.
2. TMLE begins with flexible estimates of $Q(A, W)$ and $g(W)$.
3. The clever covariate and logistic fluctuation target Q for the selected parameter.
4. Averaging $Q^*(1, W) - Q^*(0, W)$ gives a substitution estimate.
5. The efficient influence function supports the standard error and provides an estimating-equation diagnostic.
6. Double robustness and machine learning improve estimation under stated conditions; the causal interpretation retains the study assumptions.

Reproducible result

$\widehat{\Psi}_{\text{TMLE}} = -3.05$ percentage points, 95% CI $[-4.60, -1.50]$, in synthetic teaching data.

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References: Machine Learning and Inference

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